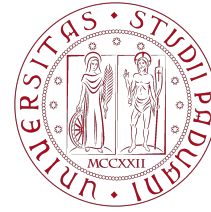


Final Report

Physics of Complex Networks: Structure and Dynamics



UNIVERSITÀ
DEGLI STUDI
DI PADOVA

Areas of physics by complexity



Newton's
Mechanics

Electro-
Magnetism

Special
Relativity

Quantum Mechanics
General Relativity

Quantum
Field Theory

Complexity
Science

Epidemics on temporal networks & the AS-level Internet from BGP

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Contents

1	Epidemic spreading on temporal networks	1
1.1	Models, topologies and mean-field thresholds	1
1.2	Results	2
1.3	Supplementary material	3
2	Monthly snapshots of the Internet AS-level network from BGP	4
2.1	Data and network reconstruction	4
2.2	Results	4
2.3	Supplementary material	6
3	Bibliography	9

1 | Epidemic spreading on temporal networks

Task leader(s): *Giancarlo Venturato*

1.1 | Models, topologies and mean-field thresholds

We study the four canonical compartmental models on *time-varying* contact networks: SI, SIS (I→S), SIR (I→R) and SEIR (S→E→I→R), all run by one discrete-time engine [8]. On every contact of snapshot t an S–I pair transmits with probability β ; an infectious node recovers with probability μ (and an exposed one matures with probability σ in SEIR). The order parameter is the endemic prevalence (SI/SIS) or the final epidemic size (SIR/SEIR), non-zero only above a critical ratio $(\beta/\mu)_c$. On an *activity-driven* (AD) temporal network the heterogeneous mean-field threshold is set by the *activity* moments, not by the aggregated degree [9], whereas compressing the observation window into a single static graph gives the classical heterogeneous (HMF) and quenched (QMF) mean-field values evaluated on that aggregate:

$$\left(\frac{\beta}{\mu}\right)_c^{\text{AD}} = \frac{1}{m(\langle a \rangle + \sqrt{\langle a^2 \rangle})}, \quad \left(\frac{\beta}{\mu}\right)_c^{\text{HMF}} = \frac{\langle k \rangle}{\langle k^2 \rangle}, \quad \left(\frac{\beta}{\mu}\right)_c^{\text{QMF}} = \frac{1}{\lambda_{\max}(A)}. \quad (1.1)$$

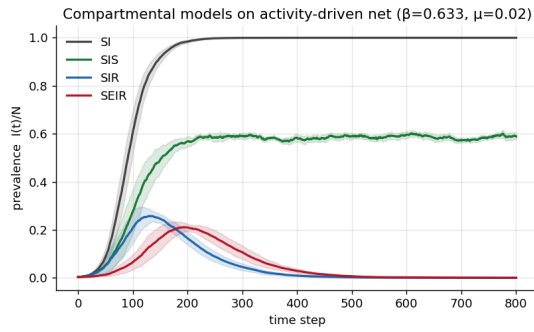
Because the aggregate accumulates every contact ever made, $\langle k^2 \rangle$ and $\lambda_{\max}$ are large and the static thresholds sit orders of magnitude below the temporal one: assuming contacts to be simultaneous systematically over-estimates spreading. Quantifying this contrast is the aim of the study.¹

The models are exercised across a variety of topologies (Holme & Saramäki [4]): (i) the AD model [9], where node i is active with probability $a_i \propto x^{-\gamma}$ per step and fires m random links (no fixed backbone); (ii) an *edge-activated* process on a fixed scale-free (Barabási–Albert) or homogeneous (Erdős–Rényi) backbone whose edges switch on with probability p per step; and (iii) the empirical *SocioPatterns Hypertext 2009* face-to-face contact network [5] (113 attendees, ~ 59 h, 5-min snapshots). Every temporal run is compared with two null models built from the same data — the *aggregated static* network and a *time-shuffled* surrogate (contacts reassigned to random steps, preserving the aggregate and contact volume) — with results averaged over 8–40 seeded realisations (supplementary Sec. 1.3).

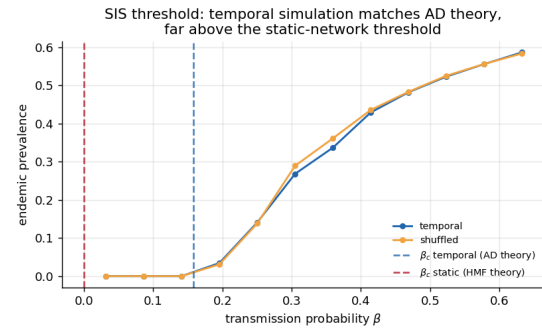
¹The simulator uses discrete-time per-step probabilities (β, μ) , which map onto the continuous rates of Eq. (1.1) for $\beta, \mu \ll 1$; the agreement in Fig. 1.1(b) confirms the mapping in the regime studied.

1.2 Results

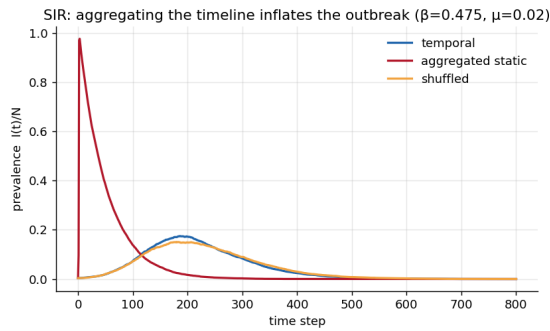
Figure 1.1(a) shows the four models on one AD network above threshold: SI saturates, SIS relaxes to an endemic plateau, and SIR/SEIR are single outbreaks, with SEIR delayed by the latent stage. Panel (b) locates the SIS threshold: the simulated prevalence departs from zero exactly at the AD prediction of Eq. (1.1) — $\beta_c^{\text{AD}} = 0.158$ versus $\beta_c^{\text{HMF}} = 1.9 \times 10^{-4}$ at $\mu = 0.02$, a factor ~ 800 (Table 1.1). The time-shuffled curve coincides with the temporal one: in a memoryless AD model the *ordering* of contacts is immaterial — what matters is concurrency. Panel (c) isolates that effect for SIR: aggregating the timeline turns a slow, bounded outbreak (peak prevalence ≈ 0.17) into an explosive one (peak ≈ 0.98 , final size 1). Panel (d) varies the backbone *structure*: at equal $\langle k \rangle = 4$, the scale-free backbone spreads markedly more than the homogeneous one (peak 0.12 vs. 0.06), both far below their common aggregated-static limit. On the empirical contact network the picture repeats — aggregation inflates the final size from 0.20 to 0.999 — with one instructive difference: *time-shuffling now lowers the outbreak* ($0.11 < 0.20$), because real contacts carry temporal correlations (bursts, group co-presence) that memoryless models lack (supplementary Fig. 1.2). Both the timing *and* the ordering of contacts matter in real data.



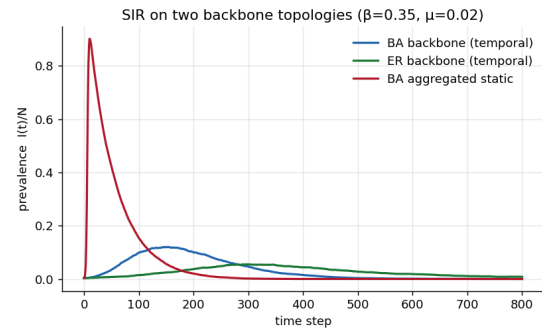
(a) Four models on the AD network



(b) SIS threshold vs. theory



(c) SIR: temporal vs. aggregated (AD)



(d) SIR: BA vs. ER backbones

Figure 1.1: Epidemics on temporal networks. (a) SI/SIS/SIR/SEIR prevalence on one activity-driven network ($\beta = 4\beta_c^{\text{AD}}$); bands are $\pm 1\sigma$ over realisations. (b) SIS endemic prevalence vs. β : the temporal simulation matches β_c^{AD} (dashed) and lies far above β_c^{HMF} ; time-shuffled $\approx$ temporal. (c) Aggregating the timeline (red) inflates the SIR outbreak relative to the temporal dynamics (blue). (d) Backbone structure at fixed $\langle k \rangle$: the scale-free BA network spreads more than the homogeneous ER one.

1.3 | Supplementary material

Method details. Synthetic defaults: $N = 1500$, $T = 800$ steps, $m = 1$, $\gamma = 2.1$, $\varepsilon = 10^{-2}$, $\mu = 0.02$. The AD activity distribution is $a_i \propto x^{-\gamma}$ with $x \in [\varepsilon, 1]$; backbone edge-activation uses p chosen to give a contact volume comparable to the AD runs. The empirical dataset (SocioPatterns Hypertext 2009, 20-s resolution) is binned into 5-minute snapshots. All experiments are seeded and reproducible with a single command; generators, the epidemic engine, threshold calculators and the output edge lists (aggregated networks in `node_from,node_to,weight` format) are in `code/temporal_epidemics/` and `data/temporal_epidemics/`.

	β_c^{AD} (temporal)	β_c^{HMF} (static)	β_c^{QMF} (static)
$(\beta/\mu)_c$	7.91	9.5×10^{-3}	1.1×10^{-2}
β_c (at $\mu = 0.02$)	0.158	1.9×10^{-4}	2.2×10^{-4}

Table 1.1: Critical points for the default AD network ($\langle a \rangle = 0.041$, $\langle a^2 \rangle = 7.4 \times 10^{-3}$; aggregate $\langle k \rangle = 61$, $\langle k^2 \rangle = 6.4 \times 10^3$). The temporal threshold is $\sim 800\times$ the static one.

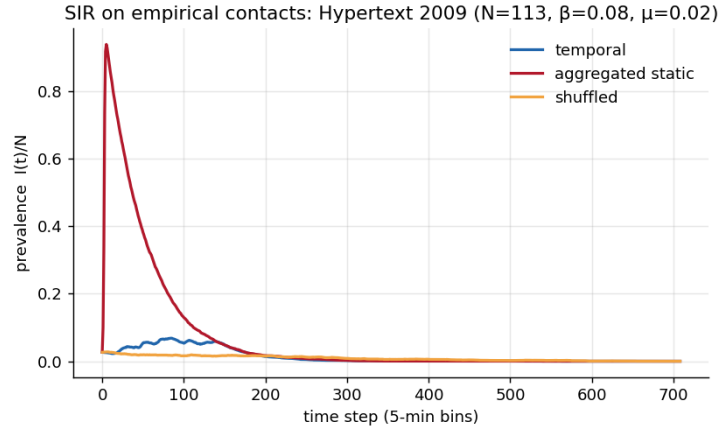


Figure 1.2: Application to empirical data: SIR on the SocioPatterns Hypertext 2009 face-to-face contact network. Aggregating the timeline (red) inflates the outbreak from a final size of 0.20 (temporal) to 0.999; unlike the memoryless synthetic models, time-shuffling (orange) *reduces* spreading (0.11), revealing temporal correlations in the real contacts.

2 | Monthly snapshots of the Internet AS-level network from BGP

Task leader(s): *Giancarlo Venturato*

2.1 | Data and network reconstruction

At the coarsest scale the Internet is a network of *Autonomous Systems* (ASes) exchanging reachability information through the Border Gateway Protocol (BGP). We reconstruct one AS-level snapshot per month from the routing-table dumps of the RIPE RIS collector `rrc00` [10], taking each month’s first available `bview` (day 01 at 00:00 UTC when present; the exact file used is recorded in a manifest). The archive spans **November 1999 – April 2026**; 314 of the 318 calendar months yield a usable table (April–July 2000 are missing from the archive; truncated files are replaced by the month’s largest dump). Three archive formats occur — modern MRT (parsed with `bgpdump`), pre-2001 Zebra MRT and the GateD text tables of 2000, the latter two read by purpose-written parsers (supplementary Sec. 2.3).

Every `AS_PATH` is cleaned: consecutive duplicate ASNs are collapsed (route prepending); malformed or empty paths, paths containing `AS_SETs` (median 0.02% of entries per month) and paths shorter than two ASNs are dropped. At least 96.4% of RIB entries are retained in every month (median 99.9%). Each consecutive ASN pair of a retained path becomes an undirected edge in canonical (min, max) order (self-loops discarded); the integer weight counts the RIB entries in which the pair appears that month — an observation frequency, *not* capacity, bandwidth, traffic volume or economic relationship. Ignoring weights yields the unweighted simple graph used for all metrics. ASNs are the integer AS numbers from the dumps, so node identity is consistent across snapshots; per-month cleaning statistics accompany the edge lists.

An essential caveat: these graphs are *not* the physical Internet topology but the AS-level topology *as observed from a single collector*. An edge is visible only if some BGP peer of `rrc00` propagates a route through it, so peripheral (especially peer-to-peer) links are systematically under-sampled and all counts are lower bounds [6]. Moreover the collector’s peer set grows in time (4×10^5 RIB entries per table in 1999, 5.5×10^7 in 2026), so part of any trend reflects improving *visibility* rather than Internet structure; we flag below where this matters.

2.2 | Results

Figure 2.1 summarises 26.5 years of evolution: the visible AS graph grows from 6 045 nodes and 10 634 edges (Nov. 1999) to 86 114 nodes and 239 705 edges (Apr. 2026),

i.e. $\times 14.2$ and $\times 22.5$. Densification is mild — $\langle k \rangle$ rises from 3.5 to 5.6 — and the sharp 2019–2021 step coincides with a major expansion of the collector’s peer set (a visibility effect). The graph is effectively a single connected component throughout (LCC $\geq 99.99\%$ of nodes in every month; two snapshots are drawn in supplementary Fig. 2.5).

The degree distribution is heavy-tailed in every snapshot (supplementary Fig. 2.2), the classical signature of the AS graph [3]. Discrete maximum-likelihood fits [2, 1] give a strikingly stable exponent, $\hat{\alpha} \in [2.01, 2.21]$ across the whole series (typical $\hat{\sigma} \approx 0.025$), averaging 2.12 in the first five years and 2.10 in the last five: the scale-free character of the AS topology persists while the network grows by more than an order of magnitude. A log-likelihood comparison with the lognormal [2] is inconclusive in 280/314 months ($p > 0.1$): “power law” here describes the regime $P(k) \sim k^{-2.1}$ over ~ 3 decades, not a model-selection claim.

Beneath the smooth growth the membership churns (supplementary Fig. 2.3): in a median month 1.3% of ASes are new and 0.7% disappear (node Jaccard overlap 0.98); edges are far more volatile (Jaccard 0.90), through genuine rewiring and fluctuating route visibility. The apex is instead rigid: consecutive months share 95% of their top-20 highest-degree ASes, and the dominant transit providers change only on multi-year timescales — the 1999 leaders (AS701 UUNET, AS1239 Sprint) give way to AS174 (Cogent), AS3356 (Level3/Lumen) and, since the mid-2010s, AS6939 (Hurricane Electric); hub *identity* is sticky even as hub degrees grow by roughly an order of magnitude (and up to ~ 100 -fold for AS174 and AS3356 relative to their 1999 values). Degree–degree correlations are strongly disassortative throughout, $\langle k_{nn} \rangle(k) \sim k^\mu$ with $\mu \approx -0.53$ around 2000 [7], drifting steadily to $\mu \approx -0.70$ by 2024–2026: low-degree customers hang ever more exclusively off high-degree providers. Leiden community detection [11] finds significant but evolving mesoscale structure: modularity rises to a plateau of $Q \approx 0.72$ around 2010, then falls to ≈ 0.61 after 2019 while the number of communities more than doubles (~ 30 to ~ 80): more visible interconnection (IXPs, direct content peering) flattens the hierarchy, and a better-instrumented collector sees more of it (supplementary Fig. 2.4).

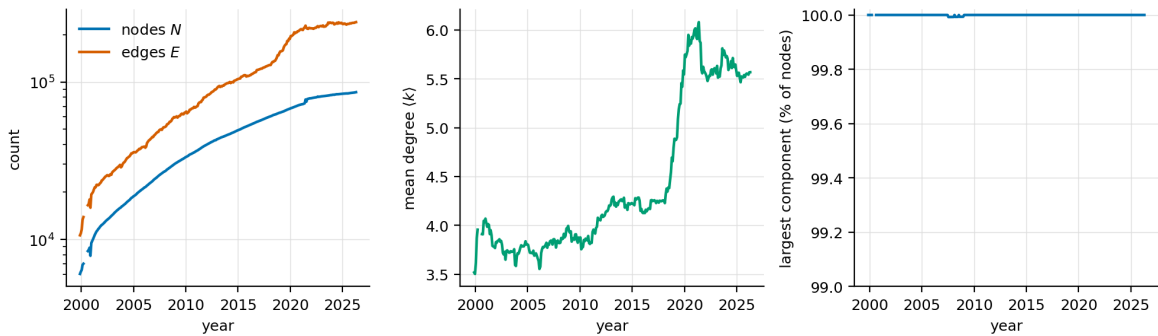


Figure 2.1: Growth of the monthly AS graph. Node/edge counts (log scale), mean degree (the 2019–2021 step is largely increased collector visibility) and giant-component share; line breaks mark the 2000 archive gap.

2.3 | Supplementary material

Pipeline details. One dump per month is downloaded and parsed in a streaming fashion (raw dumps reach ~ 1 GB in 2020–2022 and are deleted after parsing; the manifest makes the exact inputs reproducible). Truncated or corrupt archive files are detected by requiring $\geq 5 \times 10^4$ usable AS_PATHs, and the pipeline then retries with the remaining files of the month, largest first. The two legacy formats were decoded from the raw bytes and validated by checking that node counts, degree exponents and k_{nn} correlations join smoothly with the months parsed by `bgpdump`. Scripts (`build_edges.py`, `analyze_networks.py`, `make_figures.py`), the monthly edge lists, the manifest and the per-month cleaning statistics are in `code/internet_as_bgp/` and `data/internet_as_bgp/`.

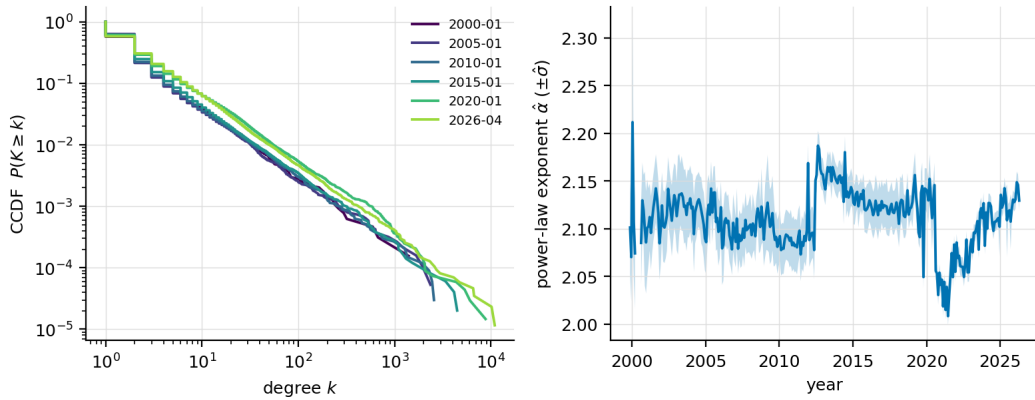


Figure 2.2: (a) Degree CCDF of six snapshots spanning the series: the tail $P(K \geq k) \sim k^{-1.1}$ ($\alpha \approx 2.1$) simply extends to larger k as the network grows. (b) Discrete MLE power-law exponent per month, with the fitted $\pm \hat{\sigma}$ band.

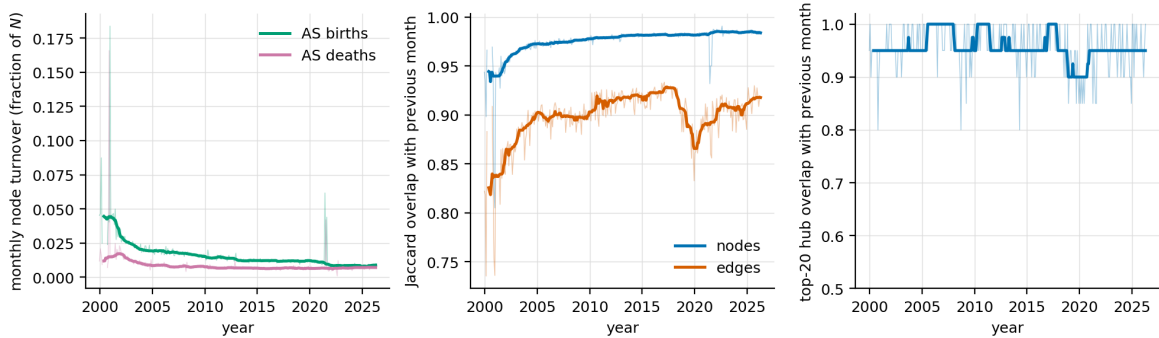


Figure 2.3: Monthly turnover (thin: raw; thick: 12-month rolling median). (a) AS birth/death rates. (b) Node and edge Jaccard overlap between consecutive months. (c) Top-20 hub set overlap between consecutive months.

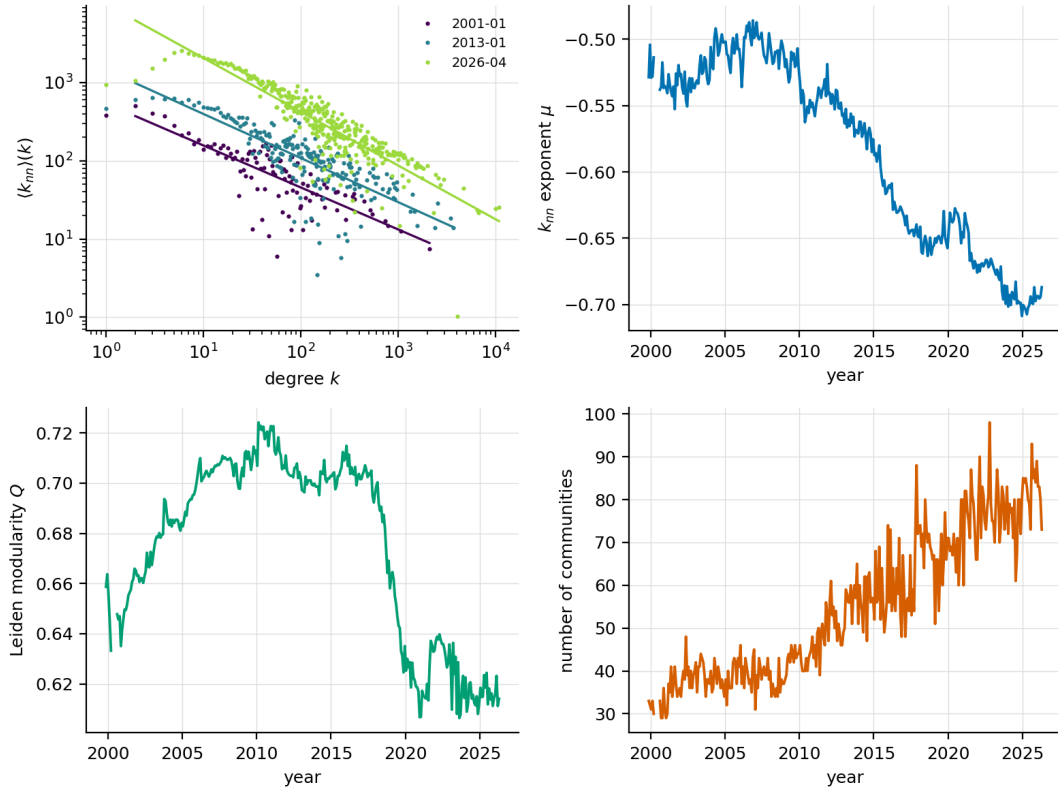


Figure 2.4: (a) $\langle k_{nn} \rangle(k)$ for three snapshots with power-law fits. (b) Disassortativity exponent μ per month. (c) Leiden modularity and (d) number of communities per month.

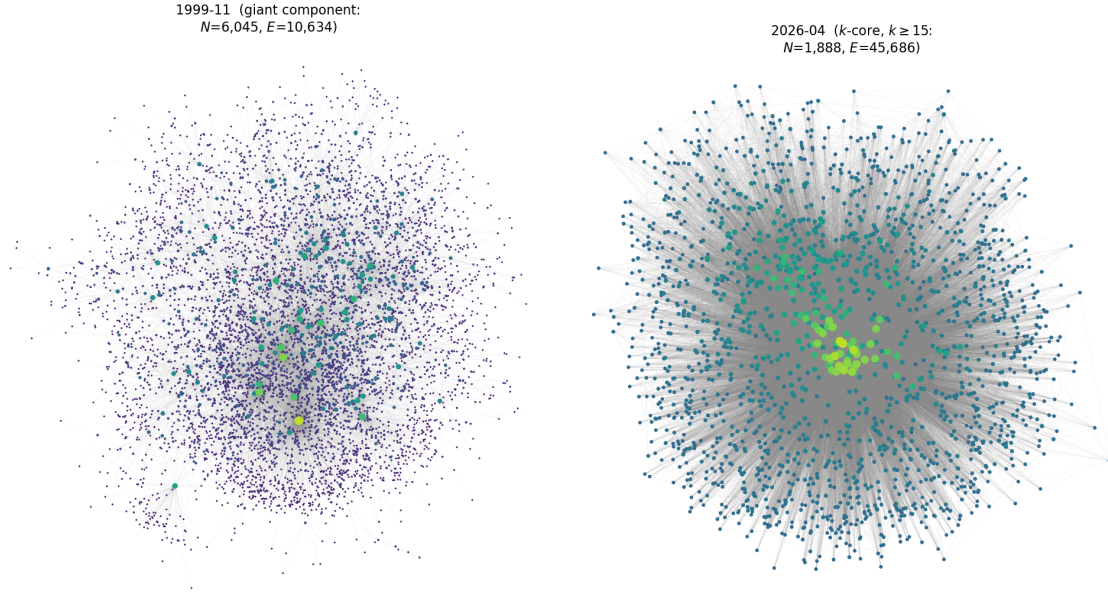


Figure 2.5: Network drawings (Fruchterman–Reingold layout; node size and colour grow with $\log k$). Left: the full giant component of the first snapshot (Nov. 1999). Right: the inner k -core ($k \geq 15$) of the last snapshot (Apr. 2026), whose 1 888 ASes concentrate the transit backbone — a bright high-degree nucleus surrounded by progressively lower-degree layers, the core–periphery organisation implied by the disassortative $\langle k_{nn} \rangle(k)$.

3 | Bibliography

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